**2** A plane is flying with a velocity of 240 m $s^{-1}$ at an angle of 30° with respect to the horizontal, as shown in Fig. 2.1. At an altitude of 2.0 km, a projectile is released from the plane. The projectile hits the target on the ground. Assume air resistance is negligible.

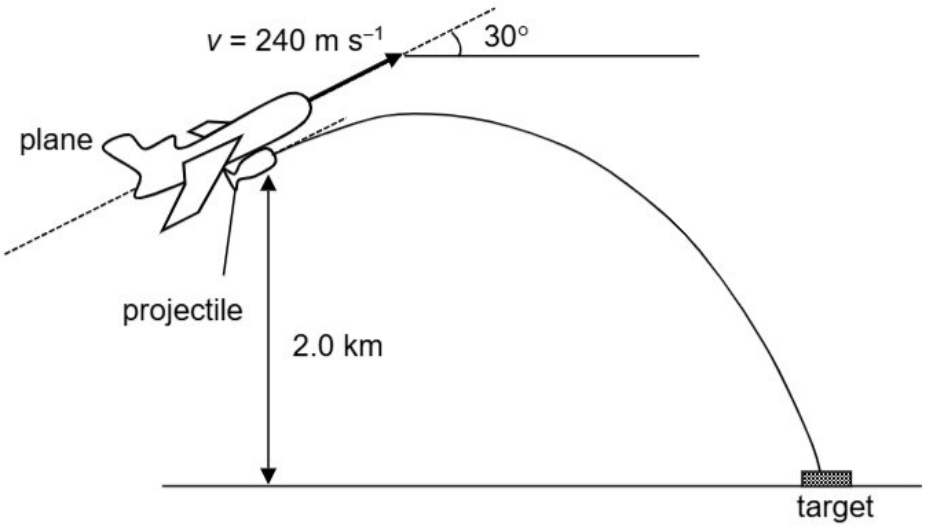



**Fig. 2.1**

**(a)** Determine the horizontal displacement of the projectile from the point of release to the target.

horizontal displacement = ..... m [3]

**(b)** Determine the speed of the projectile just before it hits the target.

speed = ..... m $s^{-1}$ [2]

**(c) (i)** Without further calculation, on Fig. 2.2, sketch the variation of vertical velocity $v_y$ with time $t$ of the projectile from the time it was released to the time it hits the target. Label this line **P**. Include values on your axes. [1]

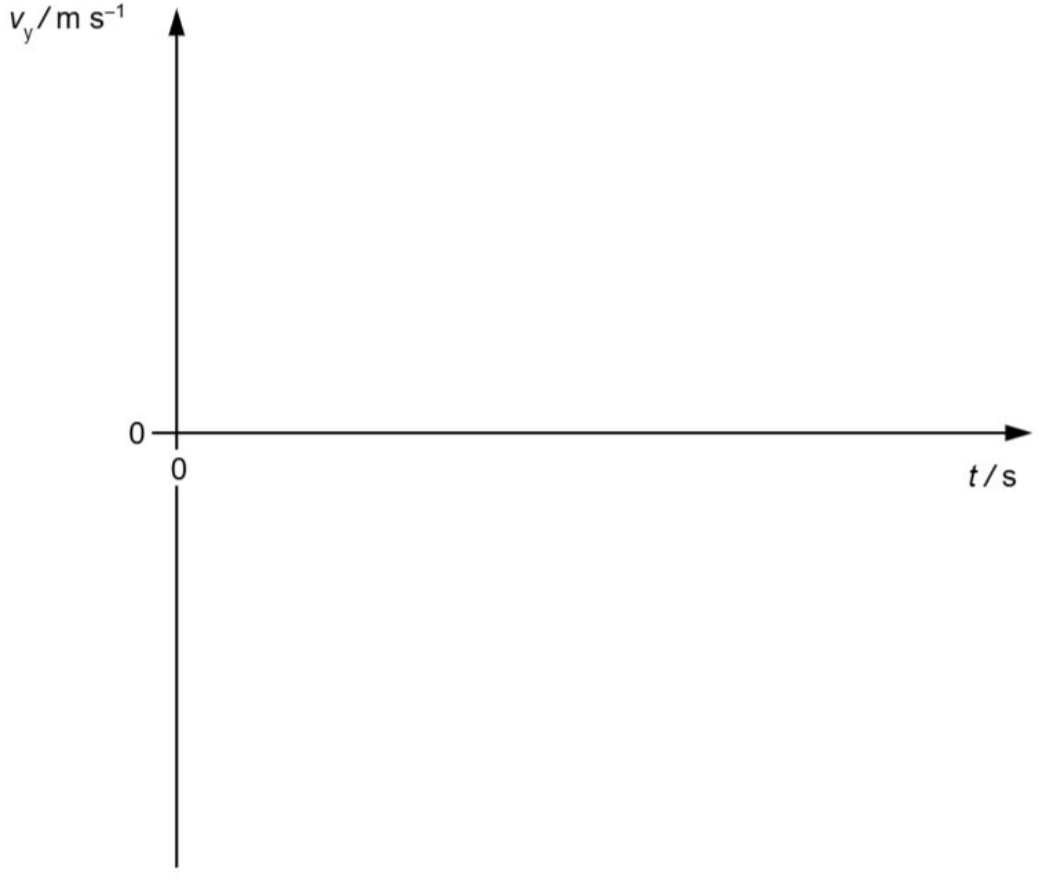



**Fig. 2.2**

**(ii)** On Fig. 2.2, sketch the variation of vertical velocity $v_y$ with time $t$ of the projectile if air resistance was **not** negligible. Label this line **Q**. No calculations are required. [2]

[Total: 8]